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Volume 03 — Diffusion Models, Sequence Models, and Transformers

Lectures L19-L27 · Weeks 7-9

L19: Forward Diffusion Process and Noise Schedules

Learning objectives

- Define the forward diffusion (noising) process.
- Understand variance schedules β_t .

Forward process

Given data $x_0 \sim q(x_0)$, define a Markov chain that gradually adds Gaussian noise over T steps:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

where $\{\beta_t\}_{t=1}^T$ is the **noise schedule** (typically $\beta_t \in [10^{-4}, 0.02]$).

Closed-form marginal

Define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. Then:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)$$

Equivalently:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

Intuition

As $t \rightarrow T$, $\bar{\alpha}_t \rightarrow 0$ and $x_T \approx \mathcal{N}(0, I)$ —pure noise. The forward process is fixed (no learnable parameters).

Noise schedules

Schedule	Formula	Property
Linear	β_t linear from β_1 to β_T	Simple, widely used
Cosine	$\bar{\alpha}_t = \cos^2\left(\frac{t/T+s}{1+s} \cdot \frac{\pi}{2}\right)$	Slower destruction at low t

L20: Score-Based Models and Denoising Objectives

Learning objectives

- Connect diffusion to score matching.
- Define the score function and denoising score matching.

Score function

The **score** of a distribution is the gradient of log-density:

$$s(x) = \nabla_x \log p(x)$$

Score matching objective

Train a network $s_\theta(x)$ to match the data score:

$$\mathcal{L}_{\text{SM}} = \frac{1}{2} \mathbb{E}_{p(x)} [\|s_\theta(x) - \nabla_x \log p(x)\|^2]$$

Denoising score matching

Add noise $\tilde{x} = x + \sigma\varepsilon$ and train:

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{p(x), \varepsilon} [\|s_\theta(\tilde{x}) + \varepsilon/\sigma^2\|^2]$$

Connection to diffusion

At each noise level t , the score of $q(x_t)$ is:

$$\nabla_{x_t} \log q(x_t | x_0) = -\frac{\varepsilon}{\sqrt{1 - \bar{\alpha}_t}}$$

The diffusion model learns to predict the noise ε added at step t .

L21: DDPM: Reverse Diffusion Formulation

Learning objectives

- Derive the reverse diffusion process.
- State the DDPM training objective.

Reverse process

We learn the reverse transitions:

$$p_{\theta}(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \sigma_t^2 I)$$

Simplified training objective (Ho et al., 2020)

Train a noise predictor $\varepsilon_{\theta}(x_t, t)$:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \varepsilon} [\|\varepsilon - \varepsilon_{\theta}(x_t, t)\|^2]$$

where $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$.

Sampling (reverse process)

At inference, start from $x_T \sim \mathcal{N}(0, I)$ and iterate:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_{\theta}(x_t, t) \right) + \sigma_t z$$

where $z \sim \mathcal{N}(0, I)$ for $t > 1$ and $z = 0$ for $t = 1$.

Variational lower bound

The full DDPM objective is a weighted variational lower bound on $\log p_{\theta}(x_0)$, analogous to the VAE ELBO.

L22: U-Net Architecture for Diffusion Models

Learning objectives

- Describe the U-Net architecture used in diffusion models.
- Explain skip connections and time conditioning.

U-Net structure

Input x_t (noisy) $\longrightarrow$ [Encoder blocks] $\longrightarrow$ Bottleneck $\longrightarrow$ [Decoder blocks] $\longrightarrow \hat{\varepsilon}$
 $\downarrow$ skip connections $\uparrow$

Each block: Conv $\rightarrow$ GroupNorm $\rightarrow$ SiLU $\rightarrow$ Conv, with downsampling (encoder) or upsampling (decoder).

Time conditioning

The timestep t is embedded via sinusoidal encoding and injected into each block (via AdaGN or addition to activations):

$$\text{emb}(t) = \left[\sin(t/10000^{2i/d}), \cos(t/10000^{2i/d}) \right]_{i=0}^{d/2-1}$$

Why U-Net?

- **Multi-scale features** capture both global structure and fine detail.
 - **Skip connections** preserve spatial information lost during downsampling.
 - **Time conditioning** allows a single network to denoise at all noise levels.
-

L23: DDPM Training Pipeline

Learning objectives

- Outline the end-to-end DDPM training and sampling pipeline.
- Compare DDPM with accelerated samplers.

Training algorithm

1. Sample $x_0 \sim p_{\text{data}}, t \sim \text{Uniform}(1, T), \varepsilon \sim \mathcal{N}(0, I)$.
2. Compute $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon$.
3. Predict $\hat{\varepsilon} = \varepsilon_\theta(x_t, t)$.
4. Minimize $\|\varepsilon - \hat{\varepsilon}\|^2$.

Sampling algorithm

1. $x_T \sim \mathcal{N}(0, I)$.
2. For $t = T, T - 1, \dots, 1$: compute x_{t-1} from x_t using learned reverse step.
3. Return x_0 .

Accelerated sampling

Method	Idea	Speedup
DDIM	Deterministic, non-Markovian reverse	10-50×
DPM-Solver	ODE-based solver	10-20×
Latent diffusion	Diffuse in VAE latent space (Stable Diffusion)	Memory + speed

L24: Classifier Guidance and Conditional Diffusion

Learning objectives

- Apply classifier guidance for class-conditional generation.
- Describe classifier-free guidance.

Classifier guidance (Dhariwal & Nichol, 2021)

Modify the reverse score with a pretrained classifier $p_\phi(y \mid x_t)$:

$$\hat{\varepsilon} = \varepsilon_\theta(x_t, t) - \sqrt{1 - \bar{\alpha}_t} \nabla_{x_t} \log p_\phi(y \mid x_t)$$

The gradient pushes samples toward class y .

Classifier-free guidance

Train a single conditional model $\varepsilon_\theta(x_t, t, c)$ and interpolate at sampling:

$$\hat{\varepsilon} = \varepsilon_\theta(x_t, t, \emptyset) + w \cdot (\varepsilon_\theta(x_t, t, c) - \varepsilon_\theta(x_t, t, \emptyset))$$

where $w > 1$ amplifies conditioning strength. Used in Stable Diffusion and DALL·E 2.

L25: Sequence Models, RNNs, and Backpropagation Through Time

Learning objectives

- Formulate sequence modeling with RNNs.
- Derive backpropagation through time (BPTT).

Sequence modeling problem

Given tokens $x_1, x_2, \dots, x_T$, model:

$$P(x_1, \dots, x_T) = \prod_{t=1}^T P(x_t \mid x_{<t})$$

Vanilla RNN

$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b_h), \quad y_t = W_{hy}h_t + b_y$$

BPTT

Unroll the RNN over T steps and apply standard backpropagation. Gradients flow through time:

$$\frac{\partial \mathcal{L}}{\partial h_t} = \frac{\partial \mathcal{L}}{\partial h_{t+1}} \cdot \frac{\partial h_{t+1}}{\partial h_t} + \frac{\partial \mathcal{L}_t}{\partial h_t}$$

Vanishing/exploding gradients

Repeated multiplication by W_{hh} causes gradients to vanish (long-range dependencies lost) or explode. Motivates LSTM/GRU and ultimately transformers.

L26: LSTMs, GRUs, and Sequence-to-Sequence

Learning objectives

- Explain LSTM gating mechanisms.
- Describe encoder-decoder seq2seq with attention preview.

LSTM cell

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) && \text{(forget gate)} \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) && \text{(input gate)} \\ \tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c) && \text{(candidate)} \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t && \text{(cell state)} \\ o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) && \text{(output gate)} \\ h_t &= o_t \odot \tanh(c_t) \end{aligned}$$

The cell state c_t provides a highway for gradient flow across long sequences.

Seq2seq

Encoder RNN processes input sequence into context vector; decoder RNN generates output sequence. Limitation: fixed-size context bottleneck $\rightarrow$ attention (L27).

L27: Motivation for Transformers and Attention Mechanisms

Learning objectives

- Define scaled dot-product attention.
- Motivate self-attention over recurrence.

Attention mechanism

Given query Q , keys K , values V :

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

Scaled dot-product attention equation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

The scaling factor $\sqrt{d_k}$ prevents softmax saturation for large d_k .

Self-attention

Set $Q = K = V = XW$ (projections of input sequence). Each position attends to all positions:

$$\text{SelfAttention}(X) = \text{softmax}\left(\frac{XW_Q(XW_K)^\top}{\sqrt{d_k}}\right) XW_V$$

Why transformers?

RNN limitation	Transformer advantage
Sequential computation	Parallel over sequence
Vanishing gradients	Direct connections via attention
Fixed context bottleneck	All-pairs attention

Multi-head attention preview

Apply attention h times with different projections and concatenate:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

Full transformer architecture is developed in Volume 04 (L28).

Volume 03 summary

Lecture	Core concept
L19	Forward diffusion, noise schedules
L20	Score matching
L21	DDPM reverse process
L22	U-Net for diffusion
L23	DDPM training/sampling
L24	Classifier guidance
L25	RNNs, BPTT
L26	LSTM, seq2seq
L27	Attention mechanism

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